NAME:
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First Solution:
Second Solution:

1. Suppose that discount factors for different maturities (one, two and three years) are given as 92% 85% and 79%, respectively. The Treasury issued a 2-year maturity coupon bond, paying coupons once per year with a coupon rate of 10%. The face value of the bond is $100. What will be the profit/loss to an investor who entered into a 1-year short forward contract on this bond (maturity is right before the first year coupon payment), if the expectations theory of the yield curve is correct?
a) 25 dollars profit
b) 0 dollar
c) 25 dollars loss
d) 33 dollars profit

2. It is January 24. A U.S. company expects to receive 50 million euro at the end of August. Euro futures contracts on the CME Group have delivery months of March, June, September, and December. One contract is for the delivery of 10 million euro. The company therefore takes September euro futures contracts on January 24 to hedge the original exposure. When the euro are received at the end of August, the company closes out its position. We suppose that the EURUSD futures price on January is 1.080 ($ per euro) and the spot and futures prices when the contract is closed out are 1.100 and 1.110, respectively. Calculate the gain/loss on the futures contract in dollar!
a) 1 000 000 dollars gain
b) 1 000 000 dollars loss
c) 1 500 000 dollars gain
d) 1 500 000 dollars loss